

LC Wireless Sensitive Pressure Sensors With Microstructured PDMS Dielectric Layers for Wound Monitoring

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Abstract—Mechanical pressures affect the rate and quality of dermal wound healing. It is therefore important to continuously monitor the mechanical pressure on a wound site so as to evaluate neovascularization and cellular proliferation under pressure. Based on an LC (inductor-capacitor) passive wireless principle, this paper presents the design, fabrication, and measurement of LC sensors embedded in a commercial bandage. Pyramidal polydimethylsiloxane elastomers used as the dielectric of the capacitive pressure sensor were fabricated by using a silicon anisotropic etching mold. Simulation and analysis of the LC sensor and wound environment effects were performed. The sensor embedded in the bandage stuck on the skin was measured, showing that its resonant frequency changes according to a linear function of applied pressure in a range of 0–200 mmHg with a sensitivity of -270.8 kHz/mmHg. The demonstrated LC sensor is completely flexible and wearable, which is proper for the future smart bandage.

Index Terms—Wound management, pressure monitoring, passive wireless sensor, smart bandage.

I. INTRODUCTION

IT IS known that controlled mechanical pressures on a wound site can accelerate neovascularization and cellular proliferation for improved wound healing [1], [2]. Compression therapy in clinical practice has been used as a first-line treatment using bandages for many centuries. However, the pressure bandages are difficult to apply with the required pressure and often cause pain and discomfort to the patient. Monitoring the bandage pressure would be useful in assessing whether the appropriate pressure was being applied to the wound and thus help in faster wound healing and increased patient comfort [3]. The pressure was measured by a wired connection [4], which would greatly limit the patients' mobility and interfere with their daily lives. Recent advances in wearable and flexible electronics along with wireless communication technology have made it possible to integrate the wound monitoring sensors with the bandage. Mehmood *et al.* presented a mobile-based telemetric dual mode sensing system to monitor pressure

and moisture levels in compression bandages and dressings used for chronic wound management [5]. The sensing system powered through a battery was fabricated on a 0.2 mm thick flexible printed circuit material with the pressure range being 0~60 mmHg. Farooqui *et al.* presented a wireless system for monitoring the pressure and pH values through inkjet printing on a standard bandage [6]. The monitoring system was powered by a thin lithium-polymer battery with dimensions of 2 mm × 12 mm × 12.5 mm. The pressure sensor was measured under the influence of external pressure on the bandage in the range of 5~100 mmHg. Although these sensors certainly help to monitor mechanical pressures in the wound, they often require the battery and interface circuits assembled on the bandage.

An LC wireless and passive alternative would significantly aid in monitoring the pressure effect on healing wounds efficiently for its small footprints without power sources [7]. This approach has received an increasing interest in recent years. Honda *et al.* [8] presented the bandage integrated with a capacitive touch sensor, temperature sensor, and wireless coil. The sensors were fabricated on a flexible substrate by screen printing, and the pressure effect was qualitatively characterized. Rahimi *et al.* [9] have recently proposed the LC wireless strain sensor in which a single spiral coil is only used with the parasitic capacitor as a pressure sensing capacitance. The wireless strain sensor was fabricated by defining a screen-printing mask directly over the wound dressing using laser machining, and it shows the sensor resonant frequency is a linear function of applied strain in the range of 0~35% with an average sensitivity of 150kHz/% strain. The exerted pressure commonly ranges from 14 mmHg to 40 mmHg in the bandages used for the pressure therapy, which improves the healing rates of some typical wounds such as venous leg ulcers [1]–[3]. Therefore, highly sensitive pressure sensors are desired in the bandage pressure monitoring. Microstructured elastomer PDMS for use as dielectric materials in capacitive pressure sensors has recently been demonstrated with high sensitivity and very short response times [10]–[12]. By utilizing the microstructured PDMS as the dielectric for the capacitive pressure sensor, this paper presents the design, fabrication, and measurement of the LC wireless sensors embedded in the commercial bandage. Fig.1 depicts the flexible LC passive wireless pressure sensor embedded in the commercial bandage strip acting as a cover package. In Section II, the operation

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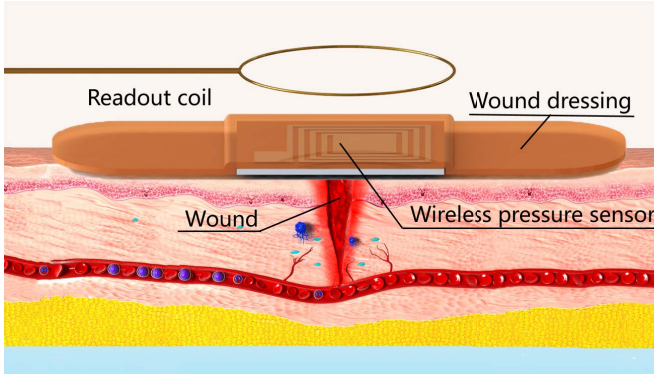


Fig. 1. Conceptual diagram of an LC wireless pressure sensor embedded in the bandage stuck on the skin wound.

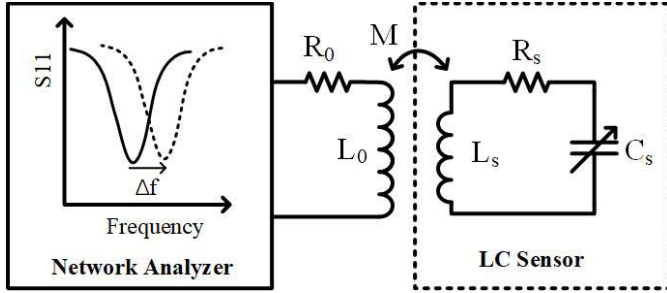


Fig. 2. Equivalent circuit of an LC passive wireless sensing system.

principle of the LC passive wireless sensor is briefly introduced. The capacitive pressure sensor with the pyramidal PDMS elastomers as the dielectric, and wound environments effect are simulated and analyzed. In Section III, a fabrication process for the flexible LC passive wireless pressure sensor is presented. In Section IV, the measurement of the fabricated device is given in a sealed gas pressure chamber. Finally, some conclusions are given in Section V.

II. DESIGN AND ANALYSIS

A. Construction of an LC Wireless Pressure Sensor

An LC sensor consists of a spiral inductor connected with a sensing capacitor, forming a resonant LC tank. As shown in Fig.1, a capacitive pressure sensor is embedded in a bandage. The capacitance changes in response to the pressure, resulting in a shift in its resonant frequency. To wirelessly interrogate the LC sensor, a readout coil is magnetically coupled with the sensor, and the resonant frequency of the sensor is detected through monitoring the impedance or input return loss of the readout coil. A schematic representation of the LC sensor is exhibited in Fig.2.

The resonant frequency (f_s) and the Q -factor (Q_s) of the LC pressure sensor can be expressed as [7]

$$f_s = \frac{1}{2\pi\sqrt{L_s C_s}} \quad (1)$$

$$Q_s = \frac{1}{R_s} \sqrt{\frac{L_s}{C_s}}, \quad (2)$$

where L_s , C_s and R_s are the inductance, capacitance and resistance of the sensor tank, respectively. The working

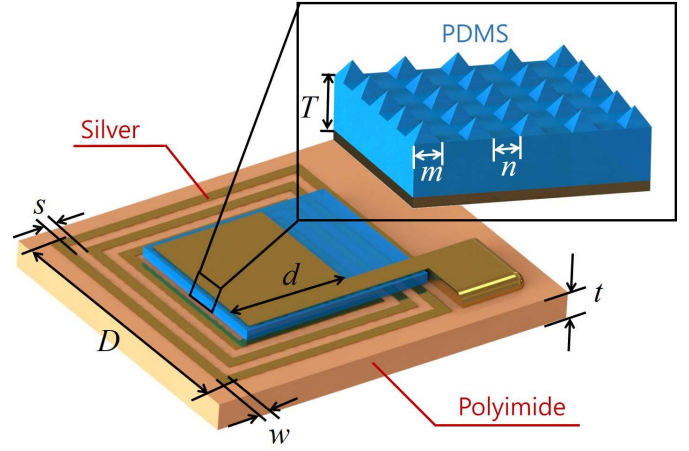


Fig. 3. Schematic diagram of a parallel plate pressure sensitive capacitor with a pyramidal PDMS layer as the dielectric.

mechanism is based on the frequency shifts caused by the variation of the sensitive capacitance in response of pressure.

Equivalent input impedance (Z_{in}) of the readout coil is written as

$$Z_{in} = R_0 + j\omega L_0 + \frac{\omega^2 M^2}{R_s + j\omega L_s - j\frac{1}{\omega C_s}}, \quad (3)$$

where ω is the angular frequency, R_0 and L_0 are the resistance and inductance of the readout coil respectively, and M is the mutual inductance between readout coil and the sensor coil. For one port measurement system, the input return loss, S_{11} parameter, can be expressed as

$$S_{11} = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} \Big|_{Z_0=50\Omega}. \quad (4)$$

where Z_0 is the characteristic impedance of the transmission line. The input return loss S_{11} is minimal near the resonant frequency of the LC sensor when the electromagnetic energy is absorbed by the sensor. Therefore, the resonant frequency of the LC passive wireless sensors is remotely detected from the minimum of the input return loss S_{11} .

B. Mechanical Simulation of Pyramidal PDMS Structures

A pressure sensor is a key component in monitoring the bandage pressure. As shown in Fig.3, an LC wireless sensor embedded in the commercial bandage is composed of a screen-printed planar spiral inductor and a parallel plate pressure sensitive capacitor with a pyramidal structured PDMS layer as the dielectric. All materials selected to fabricate the sensor and bandage are completely flexible without any inflexible electronic component. Sufficient flexibility can minimize the additional mechanical forces applied on the wound.

In order to realize the sensitivity improvements, simulation of the plane parallel capacitor with the patterned PDMS dielectric layer is carried out using the COMSOL software. The size of the pyramidal structure on the surface of the PDMS layer is listed in Table I. The PDMS material values are given in Table II. The array of the microstructure is set as 5×5 to

TABLE I
GEOMETRICAL PARAMETERS

Parameter	Description	Value
D	outer diameter of the sensor	5 mm
d	side length of the capacitor	2.6 mm
T	dielectric thickness	100 μm
t	substrate thickness	50 μm
w	conductor width	200 μm
s	conductor space	200 μm
m	microstructure size	50 μm
n	microstructure space	50 μm

TABLE II
MATERIAL PARAMETERS OF PDMS

Parameter	Description	Value
E	Young's modulus	850 kPa
ν	Poisson's ratio	0.48
ϵ_r	relative permittivity	3

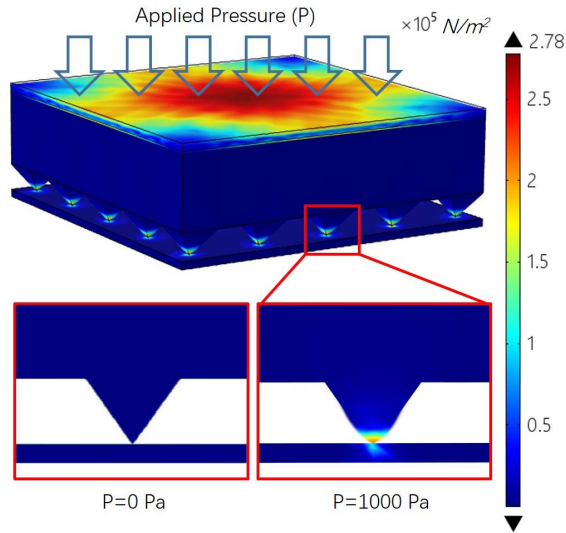


Fig. 4. Simulated stress distribution when the pressure being applied on the plane parallel capacitor is 1000 Pa.

reduce the amount of computation. The stress distribution of the capacitor with an applied pressure of 1000 Pa over the surface is shown in Fig. 4. Simulation results show an obvious compressive stress concentration in the tip of the micro pyramid, causing a large deformation. The stress concentration phenomenon can reduce the viscoelastic resistance of the microstructured PDMS dielectric layer, which is the primary factor of the sensitivity improvement. In addition to the change in separation distance, the change in effective permittivity of the region between the plates under compression is another factor. The pyramidal elastomer microstructures deform to fill in the air gaps between them with applied pressure, thus increasing the effective permittivity [11].

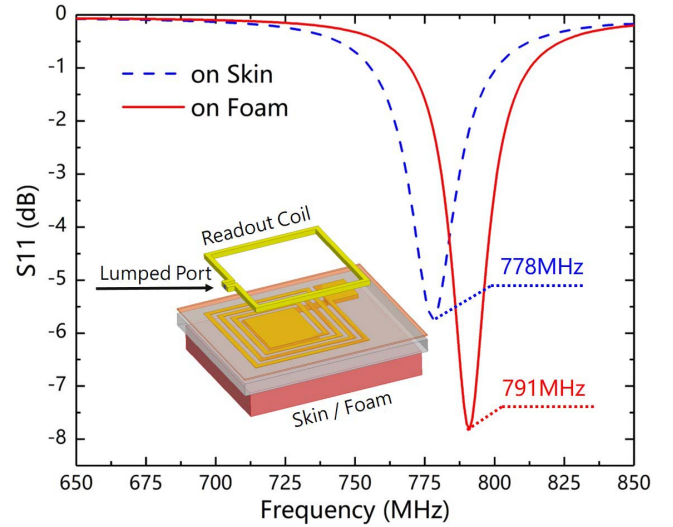


Fig. 5. Simulation results of the LC sensor in contact with the skin and the foam, respectively.

C. Wound Environment Effects

Environments usually affect an LC wireless sensor [13]. In practical applications for wound management, the performance of the pressure sensitive bandage would be affected by the wound environment. The foremost impact factor is the dielectric property of the patient's skin. The relative dielectric permittivity of the skin, air, PDMS, and polyimide is about 33, 1, 3, and 3.4, respectively. The capacitance of the sensor may be influenced when the pressure sensitive bandage is in contact with a wound bed. Here the HFSS software is used to simulate the influence of the skin. The geometrical parameters are the same as in Table 1, the outer diameter of the readout coil is 5 mm, and the distance between the sensor and readout coil is 2 mm. The distance between the pressure sensor and the skin/foam is 300 μm , which is because of the existence of the medical sponge on the commercial bandage. As shown in Fig. 5, it shows that the resonant frequency of the LC pressure sensor will be reduced by 13 MHz when the underlayment is changed from the foam to the skin, and the signal strength is also decreased. This is due to a parasitic capacitance increase when the pressure sensitive bandage is in contact with the skin. The frequency shifts caused by the skin will influence the measurement accuracy in practical applications, which will be further discussed in Section IV.

III. FABRICATION

Similar to a fabrication process for PDMS films with pyramidal feature arrays [11], the fabrication process for an LC passive wireless pressure sensor is presented in Fig. 6.

A silicon wafer was etched to produce a mould containing arrays of the inverted pyramid through anisotropic etching process. A dilute solution of the PDMS mixture of base and cross-linker (10:1) in hexane was drop cast onto the silicon wafer mould, and cured at 60 $^{\circ}\text{C}$ for 3 hours to fabricate the microstructured rubber dielectric layer. The microstructured surface of the PDMS layer is shown in Fig. 7(a). The upper

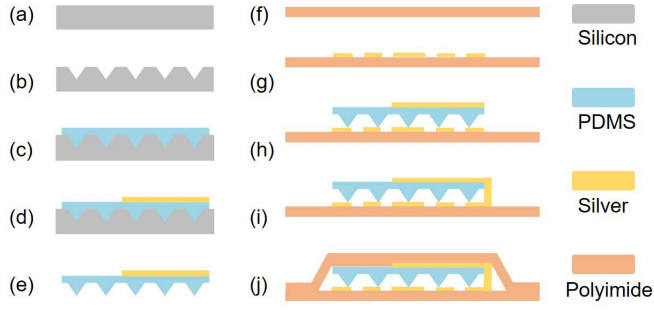


Fig. 6. Schematic process of the fabrication of the flexible LC-type passive wireless pressure sensor. (a-b) Si wafer moulded with an inverted pyramid array through anisotropic etching process. (c) Casting PDMS mixture of base and cross-linker in hexane onto the Si wafer mould and curing. (d) Printing conductive silver paste for making the top electrode of the sensor. (e) Peeling the microstructured PDMS layer off the Si mould. (f-g) Printing silver paste on the polyimide substrate for making a planer spiral inductor. (h-i) Placing the microstructured PDMS film on the polyimide substrate and connecting the upper electrode to the planer spiral inductor. (j) Packaging with the polyimide.

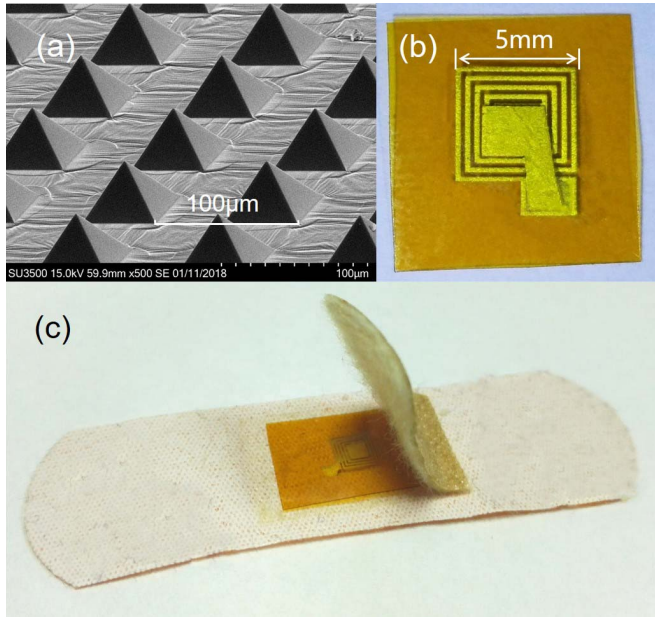


Fig. 7. (a) SEM images of the microstructured PDMS film. (b) Photograph of the fabricated LC pressure sensor. (c) Photograph of the fabricated bandage enclosed by a commercial bandage strip acting as a cover package.

electrode of the capacitor was printed on the cured PDMS layer. A planar spiral inductor was then fabricated on a 50- μm -thick polyimide film (Kapton HN) by the screen-printing technology with a screen printer (AT-25PA). The utilized conductive ink was a low-curing-temperature silver-filled polymer paste with a solid concentration of around 75%. The printed circuit was cured at 130 $^{\circ}\text{C}$ for 1 hour. The outer diameter of the inductor is 5 mm, and the line width is 200 μm . The surface microstructured PDMS layer was attached to the upper surface of the polyimide substrate, and the upper electrode was connected to the inductor using a conductive silver paste. A polyimide tape was utilized to seal the upper side of the sensor. The photograph of the fabricated LC pressure sensor is shown in Fig. 7 (b).

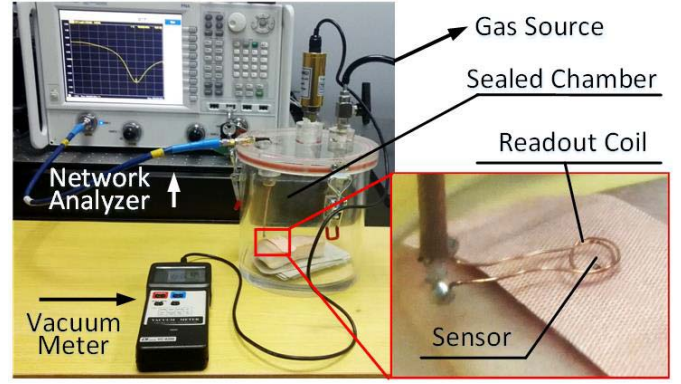


Fig. 8. Photograph of the measurement setup.

Finally, the fabricated sensor was enclosed by a commercial bandage strip utilized as a cover package to constitute the pressure sensitive bandage. The pressure sensor is between the wound dressing and the medical sponge of the bandage. As shown in Fig. 7(c), the final pressure sensitive bandage is totally flexible, which can provide a continues monitoring in the wound without negative effect on the proper function of the medical bandage.

IV. EXPERIMENTS AND RESULTS

Wireless measurement has been realized through the near-field electromagnetic coupling between an LC resonant pressure sensor and the readout coil. A 1.5-turn readout coil was connected to the network analyzer (Agilent N5224A PNA). The resonant frequency of the LC passive wireless pressure sensor was remotely monitored through the S_{11} dip measurement using the network analyzer. The distance between the readout coil and the pressure sensitive bandage is fixed at 1mm. All experiments were carried out at 25 $^{\circ}$. The measurement setup is shown in Fig. 8. In order to control the mechanical forces applied on the pressure sensitive bandage, a gas pressure chamber was utilized to test the bandage response over a broad range of pressure. The pressure source is a pump that can exhaust and inflate air in the sealed chamber so as to change the pressure inside the chamber. A vacuum meter (Tenmars VC 9200) was used to monitor the pressure change in the sealed chamber in real-time.

Since the pressure sensitive bandage is fabricated in the atmospheric environment, there are air voids existing in the region between the plates. Therefore, the practical pressure (P) applied on the bandage should be expressed as

$$P = P_{chamber} - P_0, \quad (5)$$

where $P_{chamber}$ is the gas pressure in the sealed chamber, and P_0 is the outer environmental atmosphere ($P_0 = 760 \text{ mmHg}$). In the experimental system, gas pressure in the sealed chamber can be readout from the vacuum meter directly. The pressure sensitive bandage is characterized on the surface of the foam inside the gas pressure chamber. Fig.9 shows the measured resonant frequency as a function of pressure over a pressure range of 600 mmHg. It is clear from Fig.9 that the pressure response is linear in the external pressure region of 0~200 mmHg

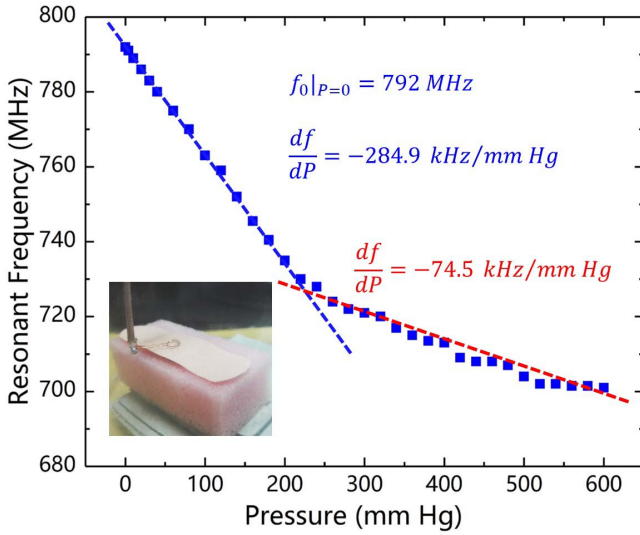


Fig. 9. Measured resonant frequency as a function of applied pressure up to 600 mmHg.

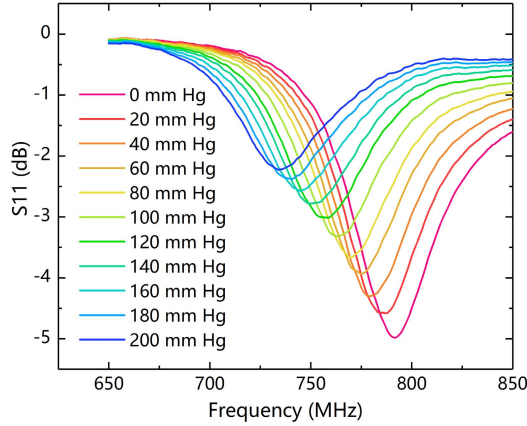


Fig. 10. Measured S_{11} parameter versus frequency with the pressure being from 0 mmHg to 200 mmHg.

with a sensitivity of -284.9 kHz/mmHg. The fitted resonant frequency (MHz) as a function of applied pressure (mmHg) is given by

$$f_s = -0.2849P + 792. \quad (6)$$

When the applied pressure is beyond about 200 mmHg, the sensor sensitivity is decreased to -74.5 kHz/mmHg, which is due to the increasing elastic resistance when the air region is fully filled, as shown in Fig.4. Since the required range of pressure monitoring for wound healing management is within 100 mmHg [5], [6], the range of 0~200 mmHg is quite enough to meet the demand of wound management.

In the high sensitive region, the increase of the capacitance with pressure will not only decrease the resonant frequency of the sensor, but also decrease the measured signal strength. As shown in Fig. 10, the S_{11} dip is decreased monotonically from 5.1 dB to 2.3 dB when applied pressure is increased from 0 to 200 mmHg. In order to ensure a readable signal with an applied pressure up to 200 mmHg, the distance between the

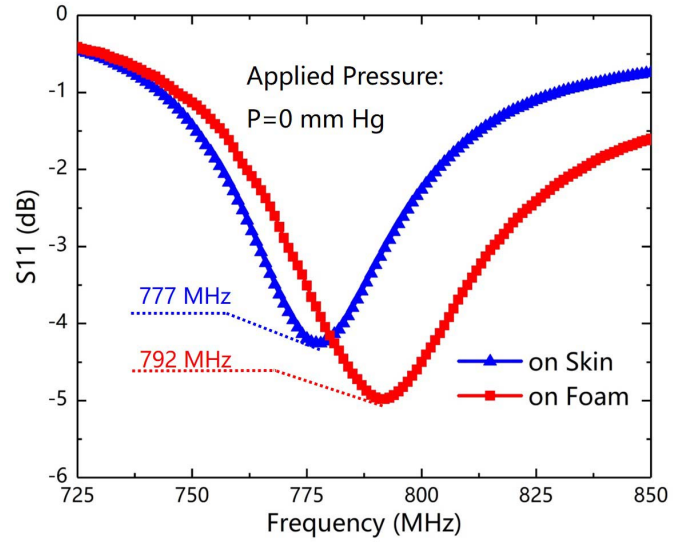


Fig. 11. Measured S_{11} parameter versus frequency without the external pressure.

readout coil and the sensor should be controlled within 3 mm. Through the measured S_{11} curves, the Q -factor of the LC pressure sensor can be calculated by

$$Q = \frac{f_s}{\Delta f_{FWHM}}, \quad (7)$$

where Δf_{FWHM} is the FWHM (full width at half maximum) bandwidth of the measured S_{11} signal. Estimated results indicate that the initial Q -factor of the sensor is 25.9, and it decreases monotonically as the capacitance increases under pressure.

In order to study the influence of the skin on the sensor performance, the sensitive bandage was stuck on the surface of the foam and the human skin, respectively. The applied external force is 0 mmHg. The measured S_{11} parameter versus frequency is shown in Fig.11. Experimental results show that the frequency decreases from 792 MHz to 777 MHz when the underlayment is changed from the foam to the skin, which matches well our simulation results, as shown in Fig.5. The frequency shift is because of the change of effective permittivity of the environment, which increases the total capacitance of the LC pressure sensor, and subsequently decreases the resonant frequency.

For a further study, a piece of fresh pork was used as the substitute of the wound skin in the gas pressure chamber to model the wound environment. Fig. 12 shows the measured resonant frequency as a function of pressure in the range of 0-200 mmHg. Experimental results show that the sensitivity of the sensor attached to the fresh pork is -270.8 kHz/mmHg, lower than -284.9 kHz/mmHg. This is because the patient's skin will not only cause the resonant frequency to decrease, but also reduce the pressure sensitivity of the bandage. The influence of the skin may be eliminated through an accurate calibration about the skin when the bandage is utilized to monitor the pressure for wound management. The calibrated resonant frequency (MHz) as a function of applied pressure

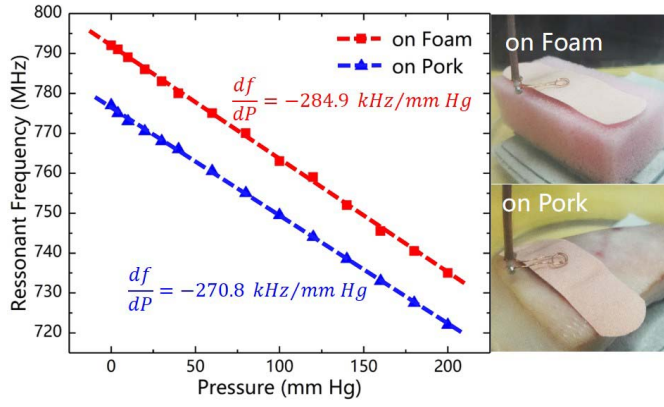


Fig. 12. Measured resonant frequency versus applied pressure up to 200 mmHg when the bandage is in contact with a pork and a foam, respectively.

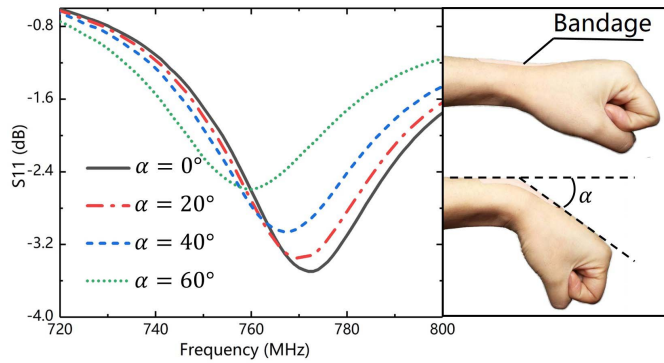


Fig. 13. Measured frequency shift versus the bending angle.

(mmHg) when the bandage is attached to the skin is given by

$$f_{S|cal} = -0.2708P + 777. \quad (8)$$

When the patient's wound is on the joint, such as knees and wrists, the patient activities will usually improve the mechanical forces on the wound surface and even tear the wound, causing a great disruption of the healing process. The pressure sensitive bandage can be used to monitor the flexion of patient joint activities through the pressure measurement. As shown in Fig. 13, the pressure sensitive bandage is attached to the patient's wrist. When the wrist bends, the resonant frequency of the sensor shifts monotonically with increasing the bending angle. When the bending angle of the wrist is changed to 60°, the measure resonant frequency is 760 MHz, which means that the pressure applied on the wound surface is increased to 62.8 mmHg, according to Eq.(8). Experimental results demonstrate that the pressure on the wound surface increases during the bending process. If the patient movements are too violent and exceed the set threshold, the sensitive bandage can be utilized to alert the patient.

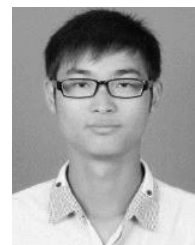
V. CONCLUSIONS

Based on an LC passive wireless principle, this paper has demonstrated the LC sensors embedded in a commercial bandage. Pyramidal PDMS elastomers used as the dielectric

of the capacitive pressure sensor have been fabricated and characterized. Simulation, analysis, and experiments of wound environment effects on the LC sensor have been performed. The sensor embedded in the bandage stuck on the skin was measured. It shows that the resonant frequency changes as a linear function of applied pressure in a range of 0–200 mmHg with a sensitivity of -270.8 kHz/mmHg. The flexible and wearable LC sensor proposed here may be used to continuously monitor the mechanical pressure on a wound site so as to evaluate neovascularization and cellular proliferation under pressure.

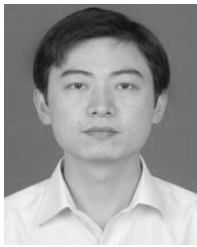
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Dr. Huang was a TPC Member of Transducers from 2009 to 2017 and the IEEE Sensors Conference from 2002 to 2015. He was a recipient of the National Outstanding Youth Science Foundation Award of China in 2003. He was the Conference Co-Chair of the SPIE Microfabrication and Micromachining Process Technology and Devices (Proceedings of SPIE, vol. 4601, in 2001), the TPC Co-Chair of the Sixth Asia-Pacific Conference of Transducers and Micro/Nano Technologies, Nanjing, in 2012. He served as the Steering Committee Chair of APCOT from 2012 to 2014. He has served as the Editor-in-Chief of the *Chinese Journal of Sensors and Actuators* since 2005 and is an Editorial Board Member of the *Journal of Micromechanics and Microengineering*.